

Document reference		NPS / 002 / 021					
Version:-	2	Date of Issue:-	Nov 09	Page	1	of	10

NPS/002/021 – Technical Specification for 33kV Power Cables

1. Purpose

The purpose of this document is to detail the technical requirements of CE Electric UK in relation to 33kV power cables.

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
NPS/002/021	1 (June 2006)	Technical Specification for 33kV Power Cables

2. Scope

This specification details the requirements for 33 kV power cables for use on CE Electric UK distribution network.

3. Technical Requirements

3.1.1 Conditions of Installation.

Cables specified in this document shall be installed in accordance with the CE Electric UK policy for the installation of cables: DSS/026. Cables specified in this document will be pulled or laid into open trenches, pulled into ducts or installed in air. Cables may also be installed directly by trenchless installation techniques. During storage and after installation cables can be expected to be subjected to the full range of climatic conditions encountered in the UK. Cable may be surrounded by ground water for most of their operating lives. Where cable is installed in ducts, flooding of ducts can occur resulting in permanently wet sections along the cable route. Cables installed above ground will be supported by means of cleats either vertically or horizontally and these cables may be exposed to direct sunlight for significant periods. Cables may be installed up wood poles in contact with the pole and therefore in contact with a pole preservation medium such as creosote.

Accessories may be cold applied or require the application of heat.

3.1.2 Conditions of Operation

Power cables purchased in accordance with this specification are required to operate under conditions stipulated in IMP/001/913. The following are general conditions of operation:

- Nominal system voltages: 33kV.
- The working voltage of any part of the system does not normally exceed the normal system voltage by more than 10%.
- Nominal system frequency: 50Hz

Document reference		NPS / 002 / 021					
Version:-	2	Date of Issue:-	Nov 09	Page	2	of	10

- The system operates with the neutral point earthed either directly or through a resistance or reactance at one or more points.

3.2 33kV Distribution Cables – Extruded Insulation

33kV distribution cables of single core extruded insulation design shall be both single core and in “triplex” configurations.

Cables shall comply in all respects with the requirements of BS 7870 - 4.10 and ENATS 09 -17 as appropriate.

All cables shall comply with the concentricity requirements of BS 7870 - 4.10 and ENATS 09 - 17 as appropriate..

3.2.1 Conductors

Conductors shall be compacted stranded copper in accordance with BS EN 60228. The range of conductor cross sections and materials required is as detailed in Appendix 1.

3.2.2 Insulation and Screens

The conductor screen, insulation and insulation screen shall be applied as a continuous single pass triple extrusion free of factory repairs. The cross linking process shall be completely “Dry Cured” and no water will be used during this process. The vulcanisation tube shall be filled with either dry nitrogen or with oil that is compatible with the insulation and screen materials.

The average thickness of the semi conducting screens shall be 1mm with a minimum of 0.6mm at any one point around the circumference. The insulation thickness shall be a minimum 8mm. Maximum permissible shrinkage shall be 2%.

The insulation screen shall be of the “fully bonded” type.

3.2.3 Metallic Screen and Moisture Blocking

All cables shall have minimum 35mm² copper wire screens and meet the required fault rating of 3kA for 2 seconds.

All cables shall be blocked against longitudinal moisture transmission around the copper wire screens, and shall pass the moisture penetration tests outlined in IEC 60840: 2004 (annexe F). The moisture content of the water blocking tape shall not exceed 50 000 ppm. Moisture blocking tape shall be provided under and over the copper wires.

3.2.4 Oversheath

The oversheath shall consist of an extruded layer of black MDPE s complying with the requirements of BS 6234 and the relevant sections of BS 7870 part 4. Maximum permissible shrinkage shall be 2%. The compound shall have a density within the range quoted in ENATS 09-17.

3.2.5 Long Term Ageing

Cable designs offered must have successfully undergone CIGRE long term ageing tests in accordance with amendment 3, clause 5.4.15 to CENELEC HD 605. See also section 3.3 and clause 4.5.

Document reference		NPS / 002 / 021					
Version:-	2	Date of Issue:-	Nov 09	Page	3	of	10

4 References

4.1 External Documentation

The cables shall comply with the relevant International Standards, British Standard Specifications and all relevant Energy Networks Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant.

Reference	Title
BS 6234	Specification for polyethylene insulation and sheath of electric cables
BS EN 60228	Conductors in Insulated cables
BS 7870	LV and MV Polymeric insulated cables for use by distribution and generation utilities
ENATS 09 - 17	Single core cable for use in substations having extruded insulation and rated voltages of 6350/11000 volt and 19000/33000 volt
ENATS 09 - 02	Specification for the supply, delivery and installation of power cables with operating voltages in the range 33kV to 400kV and associated auxiliary cables
Cenelec HD 605	Electrical Cables – Additional test methods
IEC 60840	Power Cable with Extruded Insulation and their Accessories above 30kV up to 150kV

4.2 Internal Documentation

Reference	Title
IMP/001/913	CoP for the Economic Development of the EHV System
DSS/026	Policy for the Installation of Distribution Power Cables

5 Definitions

Term	Definition
NIL	

Document reference		NPS / 002 / 021					
Version:-	2	Date of Issue:-	Nov 09	Page	4	of	10

6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	05/10/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	System Strategy	M Nicholson	21/10/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	27/10/2009

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	02/11/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	27/10/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	27/10/2009

Document reference		NPS / 002 / 021					
Version:-	2	Date of Issue:-	Nov 09	Page	5	of	10

Appendix 1 – Cable Types Utilised

33kV Cables

Single core

- 300mm² Circular Stranded Copper Conductor
- 400mm² Circular Stranded Copper Conductor
- 630mm² Circular Stranded Copper Conductor

Triplex Formation

- 300mm² Circular Stranded Copper Conductor
- 400mm² Circular Stranded Copper Conductor

Document reference		NPS / 002 / 021					
Version:-	2	Date of Issue:-	Nov 09	Page	6	of	10

Appendix 2 – Addendum to Supplier Requirements

For each power cable offered the Tenderer shall provide the following cable rating data: -

- Cable rating laid direct (Amps)
- Cable rating laid in a 150mm internal diameter duct (Amps)
- Cable rating in air (Amps)
- Maximum dc resistance per phase conductor at 20°C (Ohms/km)
- Maximum ac resistance per phase conductor at maximum conductor temperature (Ohms/km)
- Star reactance at 50Hz (Ohms/km)
- Star capacitance at 50Hz (Ohms/km)
- Charging current per phase at normal voltage and frequency (mA/m)
- Zero sequence impedance $R_0 + jX_0$ (Ohms/km)
- Minimum dynamic bending radius (mm) (In Triplex and single core formations)
- Minimum static bending radius (mm)
- Recommended pulling method and maximum pulling tension (kgF)

The following assumptions shall be made when quoting ratings: -

Single core cables are laid in touching trefoil.

Cover to top of cables being 800mm

Ground Thermal Resistivity = 1.2 K m/W

Ground temperature = 15°C

Air temperature = 25°C

The oversheath of all cables shall be marked in accordance with the requirements of the relevant Standard specified in this document.

All cables shall be metre marked throughout the length of the cable and the start and end values shall be marked on the drum label.

Cables shall be marked with some form of reference or batch number that can be used to ensure batch trace ability of materials and manufacturing facilities used in the construction of the cable.

Document reference		NPS / 002 / 021					
Version:-	2	Date of Issue:-	Nov 09	Page	7	of	10

Appendix 3 – Logistical Requirements

Drums used for 33kV cables shall have a maximum width of 1200mm and a maximum weight of 2500kg. All cable drums shall be returnable and the tenderer shall arrange to collect empty drums from the company's normal delivery locations. Tenderers shall state at the time of tender their proposed cable drum sizes and weights for each cable type offered.

All cable drums shall be marked in accordance with the relevant cable Specification or Standard. The drum label shall also contain

- (a) Customers catalogue number
- (b) Name of manufacturer
- (c) Supplied length
- (d) Rated voltage
- (e) Number of cores
- (f) Size of conductor
- (g) Type of conductor material ("Cu" or "Al")
- (h) Abbreviated description of cable construction
- (i) Gross and nett weights
- (j) Direction of rolling drum
- (k) The metre marking starts and end values
- (l) The unique reference number

Cable drums may be stored for long period's outdoors. All drum labels shall remain legible and durable under these conditions.

The ends of all cables shall be effectively sealed against the ingress of moisture by a method appropriate to the cable type. Tenderers shall detail at the time of tender their proposed sealing arrangement for each cable type offered.

The cable end projecting from the drum shall be protected from damage during transit, storage and handling on site.

The cable on the drum shall not be susceptible to damage during transit, storage and handling on site. Tenderers shall state at the time of tender their proposed method of protection for each cable

Each delivery length of cable shall be allocated a unique reference number. This number shall appear on the factory test sheet covering the cable length, shall be clearly marked on the drum on which the length is delivered and shall be referred to on all invoices and advice notes.

Document reference		NPS/002/021							
Version:-	2	Date of Issue:-	Nov - 09	Page	8	of	10		

Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION

33 kV cables required to be supplied against this specification shall comply with the latest issues of the relevant ENATS, British and International Standards specified. The following tables are intended to amplify and/or clarify the requirements of elements of these Standards but do not preclude meeting all requirements of the standards. .

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary.
- When any other code is entered the reason for non-conformance shall be entered.
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Details of the Cable Type (Voltage, Conductor Type and Size)

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each type of cable offered.

Document reference	NPS/002/021						
Version:-	2	Date of Issue:-	Nov - 09	Page	9	of	10

Specification for 33kV Power Cables				
	Clause / Requirements	Conformance Code	Remarks / Comments	
Conductor	BS EN 60228			
Conductor Screen Resistivity	ENATS 09:17 2.3.1.1 BS 7870 4:10 8.1.1			
Conductor Screen, Insulation & Insulation Screen: - Concentricity & Circularity	Continuous single pass triple extrusion (free from factory repairs)			
	Dry Cured Cross Linking			
	BS 7870 4:10 7.4 ENATS 09:17 2.1.3 and 2.1.4			
Insulation Thickness	BS 7870 4:10 4.2.3 ENATS 09:17 2.1.3			
Insulation Test	ENATS 09:17 2.3.1.2 (Table 3) BS 7870 4:10 8.1.2 (Table 6)			
Insulation Shrinkage	Max 2% (refer to table 6: ENATS 09:17 & BS 7870 4:10 for test methods).			
Insulation Screen: - Resistivity	ENATS 09:17 2.3.1.3 BS 7870 4:10 8.1.3			
	Fully Bonded			
Screen	Min 35mm ² Cu			
Longitudinal - Tapes Moisture - Moisture Blocking: Tapes : Content	Under and Over Metallic Screen Wires ≤ 50 000 PPM			
Moisture Penetration Test	IEC 60840			
Oversheath : - Tests	Extruded MDPE to BS 6234, BS 7870 Part 4 & Density as ENATS 09:17			
	Coloured Black			
	ENATS 09:17 2.3.1.4 BS 7870 4:10 8.1.4			
	Max 2% (refer to table7: BS 7870 4:10 for test methods).			

Document reference		NPS/002/021							
Version:-	2	Date of Issue:-	Nov - 09	Page	10	of	10		

Specification for 33kV Power Cables				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Compatibility Test	ENATS 09:17 2.3.1.5 BS 7870 4:10 8.1.5		
	Partial Discharge Test	ENATS 09:17 2.3.2.1 BS 7870 4:10 8.2.1		
	Bending Test	ENATS 09:17 2.3.2.2 BS 7870 4:10 8.2.2		
	Power Factor Test	ENATS 09:17 2.3.2.3 BS 7870 4:10 8.2.3 / 8.2.4		
	Heat Cycling Test	ENATS 09:17 2.3.2.4 BS 7870 4:10 8.2.5		
	Impulse Withstand Test	ENATS 09:17 2.3.2.5 BS 7870 4:10 8.2.6		
	4 Hr Voltage Test	ENATS 09:17 2.3.2.6 BS 7870 4:10 8.2.7		
	Screen Adherence @ SC	ENATS 09:17 2.3.2.7 BS 7870 4:10 8.4		
	Cold Bend Test	ENATS 09:17 2.3.2.8 BS 7870 4:10 8.1.6		
	Long Term Ageing	CIGRE / CENELEC HD 605		